Engineering Mechanics 1 · Unit_04 · UNIT 4

DRAFT — UE26ME142A · Unit_04 · UNIT 4 · CLASS 38 Moment of Inertia Numerical

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · CLASS 38 Moment of Inertia Numerical
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

CLASS 38 Moment of Inertia Numerical

1 CLASS 38 Moment of Inertia Numerical

Study focus: CLASS 38 Moment of Inertia_ Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000018 · EV0000019

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Question A slash one.

Key details

- Determine the moment of inertia of a rectangular area about the X and Y axis, and also find the polar moment of inertia about the point O.
- In this question, they have asked us to find inertia of this.
- Inertia of this rectangular lamina about this x-axis, then about the y-axis.
- After that we have to find polar moment of inertia about the point O.
- Polar moment of inertia about any point will be equal to some of the moment of inertia about the x and y axis passing through the same point, that is according to perpendicular axis.
- Then how to find inertia of this rectangular area about this axis.
- When you think of rectangle, you may remember the formula about the bottommost member as BH cube by 12 and about the centroidal x axis as bh cube by 3.
- If you look at this x-axis, it is not coinciding with the bottommost member.
- But what about is it a centroidal x-axis?
- Where will be the centroidal x-axis?
- in a rectangle.
- We know the y bar formula of a rectangle that is h by 2.
- Here what is h? h is 2 h by 3 plus h by 3, it is h itself.
- If we draw a horizontal line, this will be the centroidal x axis.
- That's why it is called as x naught.
- We don't have a standard formula.
- Then how to find inertia about this axis, x axis.

- Whenever we don't have a standard formula, we have to apply the parallel axis theorem.
- Inertia about any axis is equal to inertia about.
- Parallel centroidal axis plus area into distance square.
- This is the statement.
- Now I am considered I bar x because.
- This is horizontal, so centroidal axis also should be horizontal.
- Hence it will become I bar X.
- Next, how to find the value of I bar X then area and D.
- This is from the formulas.
- A is nothing but area of a rectangle.
- Here what is area of a rectangle?
- Base into height.
- What is this width or a base?
- It's it is B and the height is H.
- Then what is D?
- D stands for the distance between these two parallel lines.
- Nothing but X0 and X axis.
- And what is this distance?
- From here it is h by 2 minus h by 3.
- H by 2 minus h by 3 if you simplify and get it as h by 6.
- After substituting in this, simplify this, we will get the value of ixs bh cube by 9.
- This is the moment of inertia of a rectangle about the x axis.
- Next is we need to find moment of inertia of a rectangular area about the y axis.
- If you remember the formulas again, we know inertia about the leftmost member is Hbq by 1.
- Inertia about the centroidal y-axis is HBQ by 3.
- At the same time, is it a centroidal y-axis?
- This width of the rectangle is B.
- Where will be the centroidal y-axis in a rectangle?

Example or application from the lecture

- Let's start with finding Ix, that is moment of inertia of a rectangular area about the x axis.
- Let's check that.
- If you remember the basic formulas, when we consider a rectangle, inertia of this rectangle about its centroidal x axis is BH cube by 2 y.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000018 · EV0000019 · EV0000020 · +1 source units in pack.json

Core explanation · 05:00–09:59

At a distance of B by 2, if you draw a vertical line, this will be the central y axis.

Key details

- Since we don't have any formula about this y-axis, we have to apply parallel axis theorem to find inertia of this rectangular.
- lamina about the y axis.
- Inertia about any line that is i y is equal to inertia about parallel centroidal axis that is i bar y plus area into distance square.
- Always remember the statement of the parallel axis theorem.
- Next, what is this I bar Y inertia about centroidal y-axis?
- Remember the formulas.
- From the formulas we know inertia about centroidal y axis, that is this value, is HBQ by 12.
- D stands for the distance between the two parallel axes, that is y and y naught axis.
- What is the distance?
- That is b by 2 minus b by 4.
- You will get it as b by 4 itself.
- Then substitute the value of i bar y, area and distance in the equation 2 and simplify we will get the value of i y.
- The last part of this question is polar moment of inertia about the point O.
- As I said before, IZ about O is equal to Ix plus Iy.
- From the previous step Ix is BH Q by 9, substitute that here and Iy is 7 HBQ by 48.
- Once we simplify this, we will get the value of IZ as BH in.
- Ah into H square by 9 plus 7 B square by 48.
- This is the polar moment of inertia for a rectangular lamina about the point O.
- A slash 3.
- There is a narrow rectangular strip as an area of 300 mm square and its moment of inertia about the y axis is 35 to 10 power 3 mm to the power of 4.
- Open a close approximation to the polar radius of guardation about the point O.
- How to find polar radius of guardation?
- Polar radius of guaration is nothing but Kz.
- For Kz we have two formulas.
- Kz is equals to under root of IZ by A.
- That we can use when we know the polar moment of inertia about the.
- When we know the polar moment of inertia.
- This one more formula that is.
- Kz is equals to under root of kx square plus ky square.
- This formula we can use when we know the value of radius of guaration about x axis and radius of guaration about the y axis.
- Next, look at the figure.
- That is nothing but radius of guarantee the x axis.
- Since we know iy and area, we can find the value of radius of guaration about the y axis, that is ky.
- Ky is equals to under root of iy by total area A.
- Once we simplify this, we will get the value of Ky radius of variation about y-axis.
- We know radius of variation about x axis and radius of variation about y ax.

Example or application from the lecture

- Let's get into the next question.
- Before that, let's tabulate what is given in the cursing.
- The area is given as 300 mm square, consider that.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000022 · EV0000023 · EV0000024 · +1 source units in pack.json

Core explanation · 10:00–14:59

Simplify this, we will get the value of radius of garation about the point O as 18.48 mm.

Key details

- This is the final answer.
- The question is determine the ratio B by H such that Ix is equal to Iy for the area of the isoscerus triangles.
- Yet B is.
- This base of this triangle and edge is the height.
- What is Ix?
- Ix is moment of inertia about the x-axis, and what is Iy?
- I is moment of inertia of this triangle about the y axis.
- When we have a triangle, assume a right angle triangle or any other triangle.
- Moment of inertia about its base and centroidal x-axis and about the axis passing through its apex remains the same, irrespective of whether that is a right angle triangle or any other triangle like this.
- About this line is always about its base will be always BHQ by 12 about the line horizontal line passing through the apex is BHQ by 4.
- Irrespective of the triangle these two will remain the same.
- Moment of inertia this triangle about its ax, that is i will be equal to bh q by 4.
- I am going to use this formula.
- Then in this what is B?
- B is nothing but B itself, that is B by 2 plus B by 2, nothing but B, and H, what is H?
- Height of the triangle here also it is H.
- Next, I need to find IY.
- When we think of initially about y-axis, we should have only the right angle triangles.
- Here, this is one right angle triangle and this is also one more right angle triangle.
- Since these two are similar, iy will be equal to the formula is hbqed by 12.
- Next, since these two are similar, I can take it as 2 times that of 1.
- Because moment of inertia of this triangle about this line will be equal to moment of inertia of this triangle about the same line or same axis y axis.
- Next, if we have to get into the substitution part.
- Here what is h?
- H is this.
- That is nothing but H itself.
- Next, what is B?
- B stands for the base of the triangle.

- Since we have considered two triangles, the base should be taken as B by 2.
- Next, simplify this.
- Once you simplify you will get it as h bqed by forty-eight.
- Since Ix is equal to Iy, H B H cube by 4 will be equal to H B cube by 48.
- Solve it then.
- We need to find the ratio of B by H.
- The ratio will be equal to under root of 12.
- If you come across any doubts, you can post your queries in the PESU forum.

Example or application from the lecture

- Let's start with the next question A slash 4.
- And what is given here is Ix is equal to Iy.
- Try to solve the problems given in the assignment as well as in the question bank.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000026 · EV0000027 · EV0000028 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
CLASS 38 Moment of Inertia_ Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/048_CLASS_38_Moment_of_Inertia__Numerical__77ab2c76fcd84290.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 38 Moment of Inertia_ Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/048_CLASS_38_Moment_of_Inertia__Numerical__77ab2c76fcd84290.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
CLASS 38 Moment of Inertia_ Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/048_CLASS_38_Moment_of_Inertia__Numerical__77ab2c76fcd84290.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9985; only the JSON segmentation supplies the displayed timestamps.
CLASS 38 Moment of Inertia_ Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/048_CLASS_38_Moment_of_Inertia__Numerical__77ab2c76fcd84290.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.slides.md have no readable `.slides.md` text; their original page images are included.
3. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
4. 4 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
5. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
6. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
7. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
8. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00006	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare...
SRC00007	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00008	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare...
SRC00009	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare...

C Revision and retrieval practice